

# ImageCNN Evaluation Report

Generated: 10 August 2026 at 16:11

Images Tested: 700

Model Version: 1.0.0

Generation Manifest: generation\_manifest\_20260715\_095824.json

## Evaluation Summary

| Metric    | Value  |
|-----------|--------|
| Accuracy  | 0.6243 |
| Precision | 0.8946 |
| Recall    | 0.6367 |
| F1 Score  | 0.7439 |
| AUC       | 0.6053 |
| MSE       | 0.2626 |

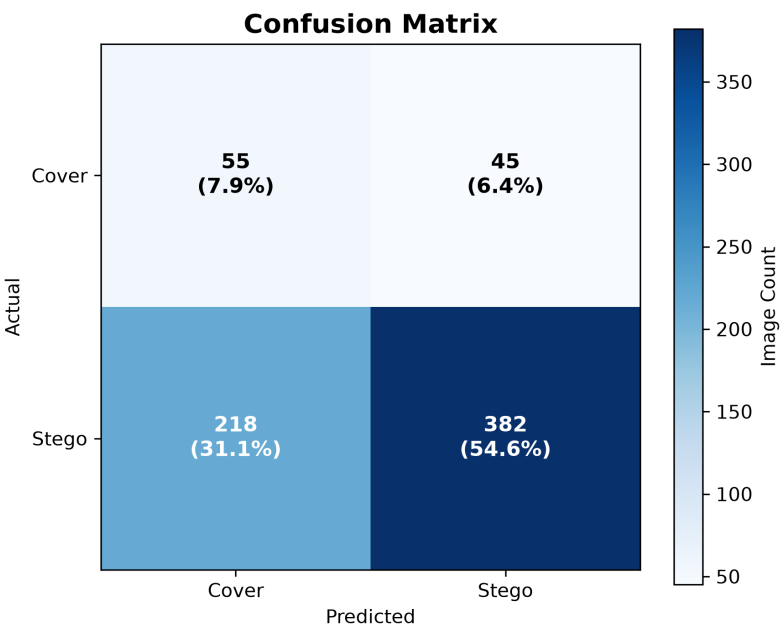
## Key Findings

- Model accuracy achieved 62.43%.
- Recall reached 63.67%, indicating successful detection of most stego images.
- Error rate was 37.57% across analysed images.
- Mean inference latency was 50.82 ms with a throughput of 19.68 images/s.

## Validation Summary

Validation reporting will be added in a future release.

# Confusion Matrix

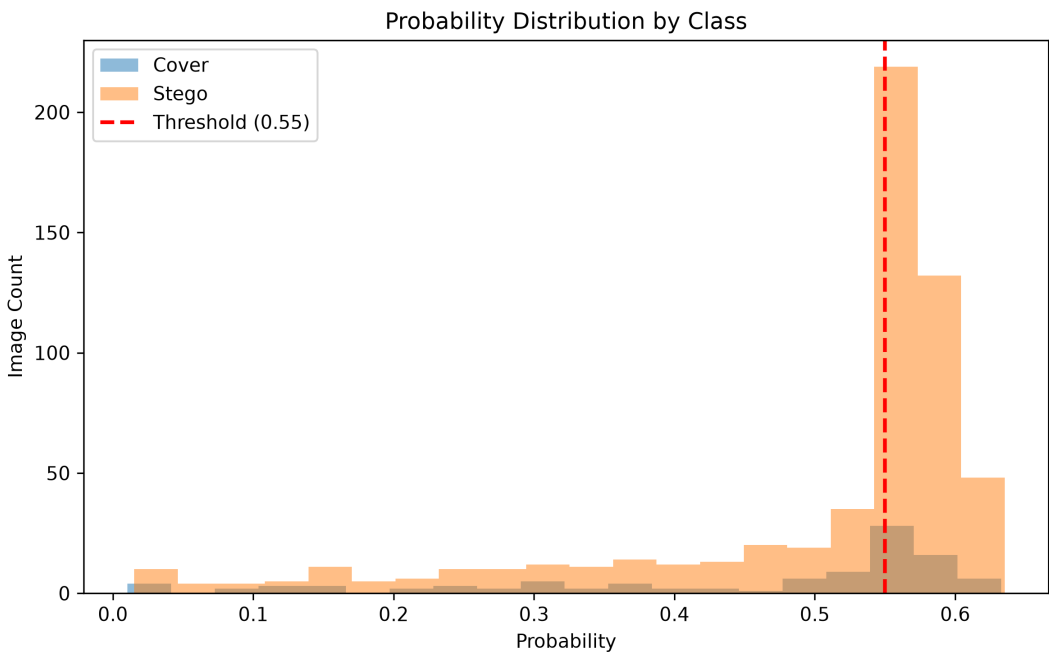


## Summary

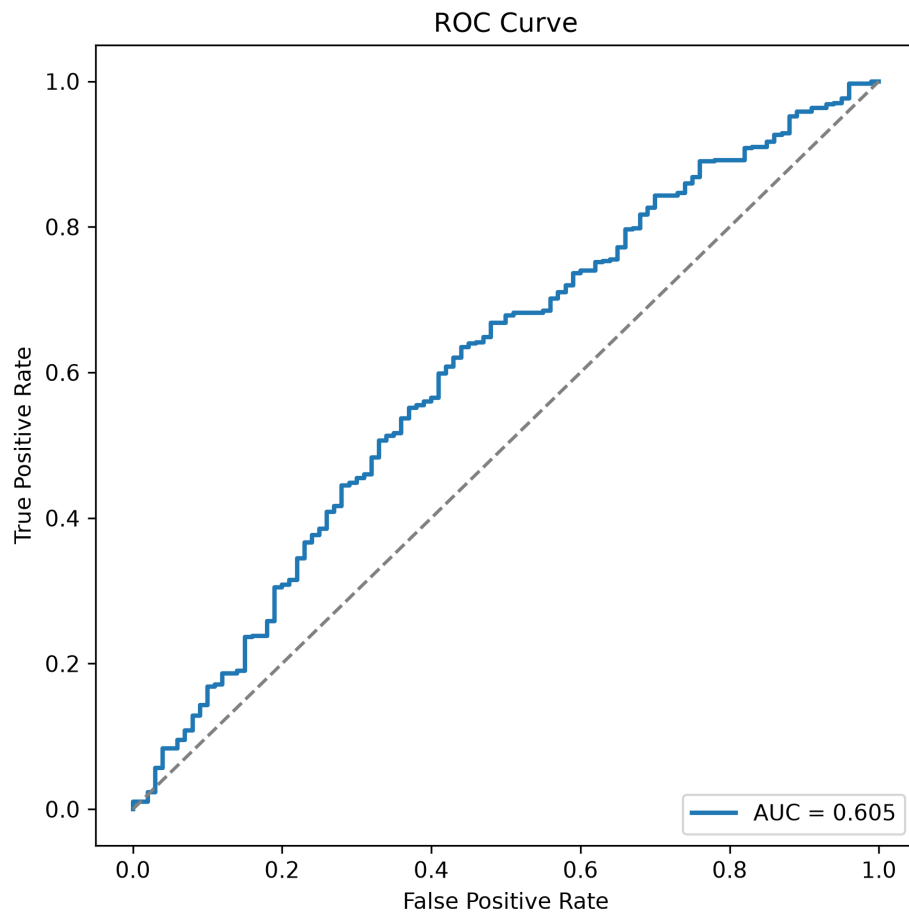
True Positives: 382  
True Negatives: 55  
False Positives: 45  
False Negatives: 218

The model correctly classified 437 images and misclassified 263 images across the evaluation dataset.

# Probability Distribution

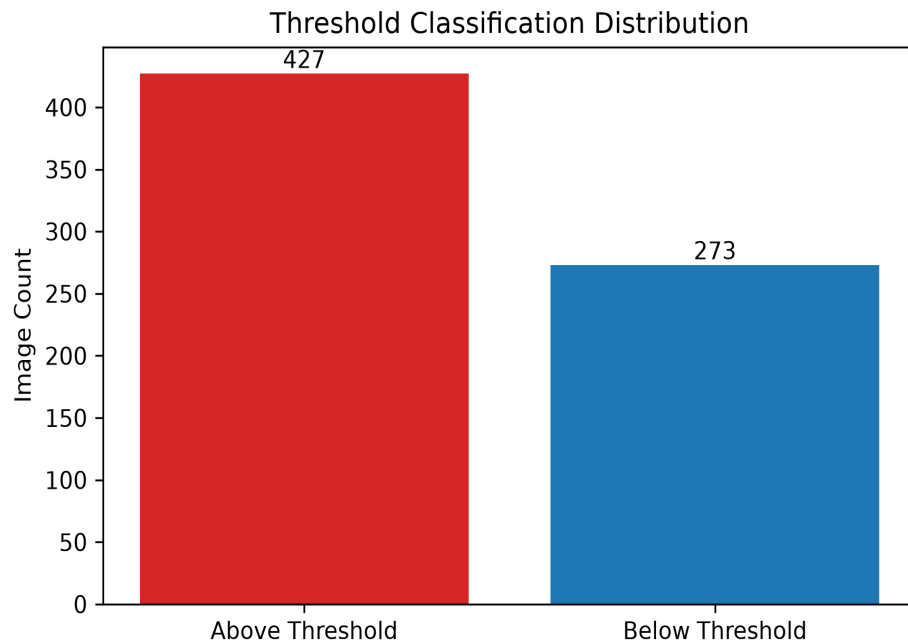


## ROC Curve



## Threshold Distribution

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### Summary

Above Threshold: 427

Below Threshold: 273

Threshold Proportion: 0.61%

0.61% of predictions exceeded the configured decision threshold.

## Classification Outcomes

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Classification outcomes summarise how predictions were distributed relative to the configured decision threshold.

Above Threshold: 427

Below Threshold: 273

Threshold Proportion: 61.00%

## Confusion Matrix Analysis

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### *Classification Results*

| Metric          | Value |
|-----------------|-------|
| Pairs Analysed  | 600   |
| True Positives  | 382   |
| False Positives | 45    |
| True Negatives  | 55    |
| False Negatives | 218   |
| Total Errors    | 263   |

### *Performance Metrics*

| Metric            | Value  |
|-------------------|--------|
| TPR (Recall)      | 63.67% |
| TNR (Specificity) | 55.00% |
| FPR               | 45.00% |
| FNR               | 36.33% |
| Error Rate        | 37.57% |

## Dataset Quality

| Folder                 | Avg PSNR | Avg SSIM |
|------------------------|----------|----------|
| <b>Stego Encrypted</b> |          |          |
| Large                  | 56.2087  | 0.998619 |
| Medium                 | 61.2381  | 0.999562 |
| Small                  | 66.5664  | 0.999871 |
| <b>Stego Plain</b>     |          |          |
| Large                  | 56.1952  | 0.998616 |
| Medium                 | 61.2893  | 0.999568 |
| Small                  | 66.5479  | 0.999870 |

*PSNR and SSIM metrics were used to assess visual similarity between cover and stego images. Higher PSNR and SSIM values indicate lower visual distortion and greater similarity to the original cover images.*

*Average PSNR and SSIM values are presented for reporting purposes. Detailed quality statistics, including minimum and maximum measurements, are provided within the accompanying workbook export.*

## Performance Analysis

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Mean Latency: 50.82 ms  
Median Latency: 47.68 ms  
Minimum Latency: 30.31 ms  
Maximum Latency: 246.03 ms  
Images Processed: 700  
Total Inference Time: 35.57 s  
Throughput: 19.68 images/s

## CNN Decision Analysis

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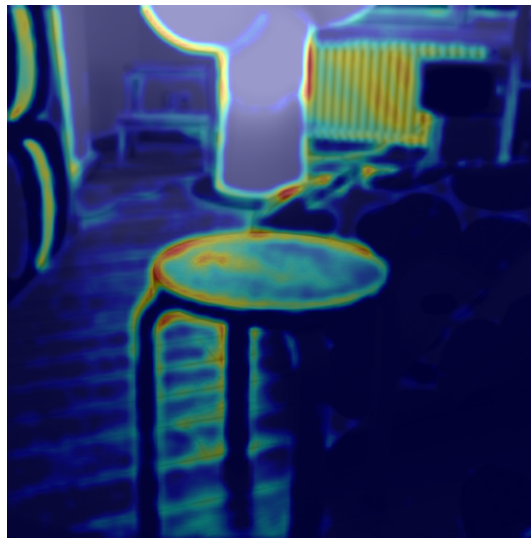
Grad-CAM activation visualisations were generated to examine the image regions that contributed most strongly to CNN classification decisions. The following analysis compares cover and stego activations, highlights areas of strongest model attention, and identifies examples exhibiting both substantial and minimal decision differences.

### ***Mean Probability Difference***

Mean probability difference shift across analysed pairs: 0.1019

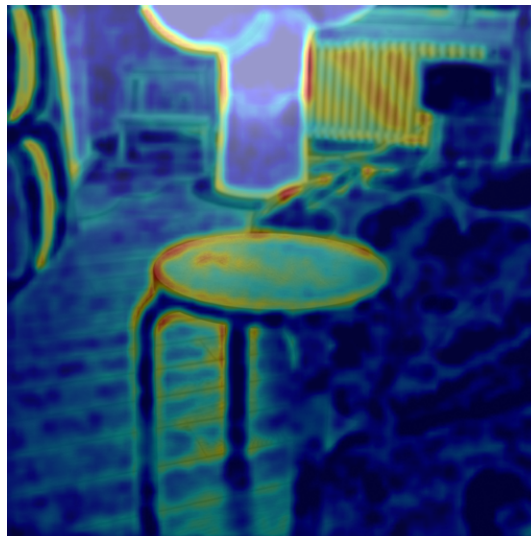
Largest difference pair: 1669 (probability difference: 0.3942)

### ***Cover Activation Map***



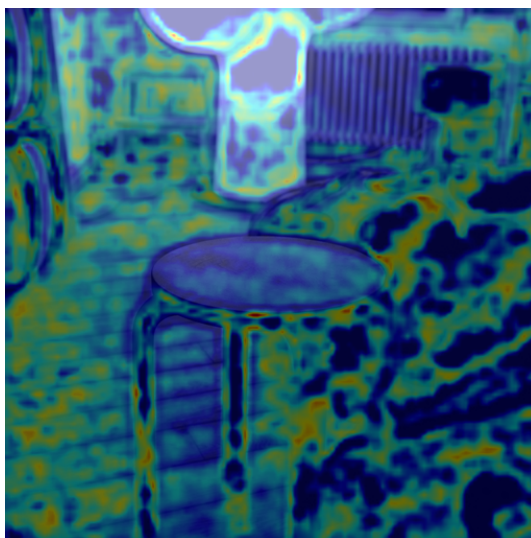
Cover Probability: 0.1290

### ***Stego Activation Map***



Stego Probability: 0.5232

## Activation Difference Map



## Analysis

A substantial probability shift of 0.3942 was observed between the cover and stego images. This suggests that features identified by the CNN were altered following payload embedding and contributed strongly to the final classification decision.

*Grad-CAM visualisation was used to identify image regions that contributed most strongly to the CNN classification decision. Highlighted regions represent influential features rather than confirmed embedding locations.*

## Most Ambiguous Pair

Image ID: 1521

Cover Probability: 0.5939

Stego Probability: 0.5915

Probability Difference: 0.0024

## Model Uncertainty Analysis

The pair '1521' produced the smallest probability difference (0.0024) between the cover and stego images and therefore represents one of the most challenging examples encountered during analysis.

## Detection Sensitivity Analysis

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### Payload Size Analysis

| Category | Accuracy | Precision | Recall | F1     | TNR    | FPR    |
|----------|----------|-----------|--------|--------|--------|--------|
| Small    | 51.67%   | 68.97%    | 50.00% | 57.97% | 55.00% | 45.00% |
| Medium   | 58.33%   | 72.73%    | 60.00% | 65.75% | 55.00% | 45.00% |
| Large    | 72.33%   | 78.26%    | 81.00% | 79.61% | 55.00% | 45.00% |

### Payload Type Analysis

| Category  | Accuracy | Precision | Recall | F1     | TNR    | FPR    |
|-----------|----------|-----------|--------|--------|--------|--------|
| Plain     | 61.50%   | 80.93%    | 63.67% | 71.27% | 55.00% | 45.00% |
| Encrypted | 61.50%   | 80.93%    | 63.67% | 71.27% | 55.00% | 45.00% |

### Payload Category Analysis

| Category         | Accuracy | Precision | Recall | F1     | TNR   | FPR   |
|------------------|----------|-----------|--------|--------|-------|-------|
| Encrypted Large  | 79.00%   | 100.00%   | 79.00% | 88.27% | 0.00% | 0.00% |
| Encrypted Medium | 62.00%   | 100.00%   | 62.00% | 76.54% | 0.00% | 0.00% |
| Encrypted Small  | 50.00%   | 100.00%   | 50.00% | 66.67% | 0.00% | 0.00% |
| Plain Large      | 83.00%   | 100.00%   | 83.00% | 90.71% | 0.00% | 0.00% |
| Plain Medium     | 58.00%   | 100.00%   | 58.00% | 73.42% | 0.00% | 0.00% |
| Plain Small      | 50.00%   | 100.00%   | 50.00% | 66.67% | 0.00% | 0.00% |

## Research Interpretation

### *Payload Size*

Large payloads achieved the strongest performance (F1: 79.61%), while Small payloads produced the weakest results (F1: 57.97%). This suggests that detection performance improves as payload volume increases.

### *Payload Type*

payload encryption had negligible impact on performance, with plain and encrypted payloads achieving comparable F1 scores (71.27% and 71.27%, respectively)

## **Research Insight**

The observed performance trend suggests that ImageCNN is responding primarily to embedding artefacts rather than payload content. Larger payloads appear considerably easier to distinguish from cover images, while encryption status has comparatively little influence on detection performance.

*Note: AUC and MSE metrics are not reported for sensitivity-analysis subsets because these subsets contain only stego samples. Both metrics require a mixture of cover and stego images to provide meaningful comparative measurements. Consequently, the sensitivity analysis focuses on accuracy, precision, recall and F1 score, which remain valid indicators of detection performance within individual payload groups.*

## Limitations

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This report provides an automated assessment based on the configured steganalysis model and evaluation parameters. Detection results represent probabilistic indicators rather than definitive proof of steganographic activity. False positives and false negatives may occur, particularly when processing previously unseen datasets, alternative embedding techniques, heavily compressed images, or images that differ significantly from the training data. Results should therefore be interpreted in conjunction with additional forensic analysis, supporting evidence, and professional judgement. Model performance is influenced by dataset characteristics, payload size, image format, calibration thresholds, and other experimental factors documented elsewhere in this report.